

Dongqi Li

M.S. Biostatistics, Northwestern University · B.A. Economics, UC Irvine

kevinldq2018@gmail.com Prospective Ph.D. Applicant, Fall 2027 [Contact](#)

This page lists publications on which I have contributed as a co-author, spanning clinical machine learning, biomedical informatics, NGS/bioinformatics workflows, and healthcare data analytics. My contributions across these collaborative projects primarily involved data cleaning and preprocessing, statistical analysis, result validation, and manuscript preparation, alongside independent modeling work on my master's thesis at Northwestern. Full author lists and venues are listed below; DOIs and IEEE Xplore links are included where available.

24

PEER-REVIEWED

1

PREPRINT

2018–2025

SPAN

Peer-Reviewed Publications (including conference papers)

2025

The Quality and Reliability of Short Videos About External Counterpulsation on TikTok: A Cross-Sectional Study

Zheng, J., **Li, D.**, Borchert, G. M., et al., & Huang, J.

Proc. IEEE International Conference on Bioinformatics and Biomedicine (BIBM 2025)

Healthcare AI

Medical Social Media

Active Learning for Meibomian Gland Segmentation in Infrared Meibography Images

Lai, K., **Li, D.**, Benton, R., et al., & Huang, J.

Proc. IEEE International Conference on Bioinformatics and Biomedicine (BIBM 2025)

2024

Machine Learning Modeling and Performance Evaluation for Gestational Diabetes Mellitus Datasets

Zhou, K., **Li, D.**, et al.

Proc. IEEE International Conference on Bioinformatics and Biomedicine (BIBM 2024), Lisbon, Portugal

Clinical ML

Diabetes

Artificial Intelligence Assisted Recognition and Diagnosis of Magnetically Controlled Capsule Gastroscopy

Chen, S., Wei, C., **Li, D.**, Borchert, G. M., Yu, Z., Lin, J., & Huang, J.

Proc. IEEE BIBM 2024, Lisbon, Portugal, pp. 5604–5609. DOI: 10.1109/BIBM62325.2024.10822068

Medical Imaging

AI Diagnosis

Relationship of Body Adipose Tissue Distribution with Vitamin D and Bone Metabolism Indices in Patients with Type 2 Diabetes Mellitus

Zhao, J., Guan, Y., Zuo, F., Nian, X., **Li, D.**, Borchert, G. M., Huang, J., & Wu, B.

Proc. IEEE BIBM 2024, Lisbon, Portugal, pp. 5670–5674. DOI: 10.1109/BIBM62325.2024.10822500

Clinical Informatics

Endocrinology

Differential Expression of Adipocyte Transcripts in SAT and VAT between Normal Weight and Obese/Overweight Individuals

Bai, C., Huang, R., Zhao, J., **Li, D.**, Borchert, G. M., Huang, J., & Wu, B.

Proc. IEEE BIBM 2024, Lisbon, Portugal, pp. 5597–5603. DOI: 10.1109/BIBM62325.2024.10822168

Transcriptomics

Metabolic Health

Evaluating the Use of Large Language Models to Identify Top Research Priorities in Hematology

Chen, H., Huang, F., Liu, X., **Li, D.**, et al., & Huang, J.

Proc. IEEE Healthcom 2023, Chongqing, China

Healthcare AI

LLM

Semi-Supervised Meibomian Gland Segmentation via Mutual Consistency Constraints and Uncertainty Rectification

Lin, J., Lu, F., Li, L., Lin, L., **Li, D.**, Borchert, G. M., & Huang, J.

Proc. IEEE BIBM 2023, eCF Paper ID: B892

Medical Imaging

Semi-Supervised Learning

2021

Utilizing Machine-Learning Techniques to Predict the Efficacy of Aerobic Exercise Intervention on Young Hypertensive Patients Based on Cardiopulmonary Exercise Testing

Huang, F., Leng, X., **Li, D.**, et al., & Huang, J.

Journal of Healthcare Engineering · PMID: indexed on PubMed

Healthcare ML

CPET

A New Classification Algorithm and a New Oversampling Method of Mapping Common Data Elements to the BRIDG Model

Li, S., Huang, Y., Kasukurthi, M. V., **Li, D.**, et al., & Huang, J.

Proc. IEEE BIBM 2021, Houston, TX

Clinical Data Standards

SURFR: A Real-Time Platform for Non-Coding RNA Fragmentation Analysis Using Wavelets

Kasukurthi, M. V., Houserova, D., Huang, Y., **Li, D.**, et al., Borchert, G. M., & Huang, J.

Proc. IEEE BIBM 2021, Houston, TX

NGS

ncRNA Analytics

2020

Development of a Support Vector Machine Learning and Smartphone IoT-Based Architecture for Real-Time Sleep Apnea Diagnosis

Ma, B., Wu, Z., Li, S., Benton, R., **Li, D.**, et al., & Huang, J.

BMC Medical Informatics and Decision Making, 20, 298

Clinical Informatics

A Partial Encryption Algorithm for Medical Images Based on Quick Response Code and Reversible Data Hiding Technology

Li, J., Zhang, Z., Li, S., Benton, R., Huang, Y., Kasukurthi, M. V., **Li, D.**, Lin, J., Borchert, G. M., Tan, S., Li, G., Ma, B., Yang, M., & Huang, J.

BMC Medical Informatics and Decision Making, 20(Suppl 14), 297

Medical Imaging

Data Security

2019

Using an Artificial Neural Network to Map Cancer Common Data Elements to the Biomedical Research Integrated Domain Group Model in a Semi-Automated Manner

Renner, R., Li, S., Huang, Y., **Li, D.**, et al., Huang, J., & Jiang, G.

BMC Medical Informatics and Decision Making, 19(Suppl 7), 276

CDE

BRIDG

Muscle Fatigue Detection and Treatment System Driven by Internet of Things

Ma, B., Li, C., Wu, Z., Huang, Y., van der Zijp-Tan, A., Tan, S., **Li, D.**, Fong, A., Basetty, C., Borchert, G. M., Benton, R., Wu, B., & Huang, J.

BMC Medical Informatics and Decision Making, 19(S7), 275

Clinical Informatics

IoT

Exploring Observability of Attractor Cycles in Boolean Networks for Biomarker Detection

Qiu, Y., Huang, Y., Tan, S., **Li, D.**, van der Zijp-Tan, A. C., Fong, A., Borchert, G. M., & Huang, J.

IEEE Access

Computational Biology

Boolean Networks

FGF21 Mediates Corticosteroid-Related Bone Mass Loss through PPAR- γ

Guan, Y., Huang, J., Huang, S., Kasukurthi, M. V., Huang, Y., **Li, D.**, Tan, S., Wang, S., Nian, X., Lin, J., Borchert, G. M., & Wu, B.

Proc. IEEE BIBM 2019, San Diego, CA

Endocrinology

Use CPET Data to Predict the Intervention Effect of Aerobic Exercise on Young Hypertensive Patients

Yang, G., Leng, X., Huang, F., Kasukurthi, M. V., Huang, Y., **Li, D.**, Lin, J., Tan, S., Lu, G., Benton, R., Borchert, G. M., Ma, B., & Huang, J.

Proc. IEEE BIBM 2019, San Diego, CA

Use Reversible Data Hiding to Protect Key Regions in Medical Images

Li, J., Zhang, Z., Li, S., Benton, R., Huang, Y., **Li, D.**, Lin, J., Tan, S., Kasukurthi, M. V., Borchert, G. M., Ma, B., & Huang, J.

Proc. IEEE BIBM 2019, San Diego, CA

Medical Imaging

Data Security

SURFr: Algorithm for Identification and Analysis of ncRNA-Derived RNAs

Kasukurthi, M. V., Zhang, D., Houserova, M., Huang, Y., **Li, D.**, Lin, J., Tan, S., Ma, B., Benton, R., Li, S., Borchert, G. M., & Huang, J.

Proc. IEEE BIBM 2019, San Diego, CA

NGS

ncRNA Analytics

2018

Mapping Common Data Elements to a Domain Model Using an Artificial Neural Network

Renner, R., Li, S., Huang, Y., Tan, S., **Li, D.**, van der Zijp-Tan, A. C., Benton, R., Borchert, G. M., Huang, J., & Jiang, G.

Proc. IEEE BIBM 2018, Madrid, Spain

Clinical Data Standards

A PWM-Based Muscle Fatigue Detection and Recovery System

Ma, B., Li, C., Wu, Z., Huang, Y., van der Zijp-Tan, A. C., Tan, S., **Li, D.**, Fong, A., Basetty, C., Borchert, G. M., & Huang, J.

Proc. IEEE BIBM 2018, Madrid, Spain

Clinical Informatics

Novel Method for Singleton and Cyclic Attractor Observability in Boolean Networks

Qiu, Y., Huang, Y., Tan, S., **Li, D.**, van der Zijp-Tan, A. C., Fong, A., Borchert, G. M., & Huang, J.

Proc. IEEE BIBM 2018, Madrid, Spain

Computational Biology

Body Composition and Biochemical Characteristics of Normal Weight Obesity in Japanese Young Women with Different Physical Activities

Huang, J., Fukuo, K., Yoshino, G., Kazumi, T., Basetty, C., Huang, Y., Tan, S., **Li, D.**, van der Zijp-Tan, A. C., Fong, A., Borchert, G. M., & Wu, B.

Proc. IEEE BIBM 2018, Madrid, Spain

Clinical Research

Preprints

SALTS — SURFR (sncRNA) and LAGOOn (lncRNA) Transcriptomics Suite

Kasukurthi, M. V., Houserova, D., Huang, Y., **Li, D.**, et al., Borchert, G. M., & Huang, J.

bioRxiv, 2021 (Preprint)

NGS

Transcriptomics Pipeline